CBC Digital Learning Platform

Stage C — Domain Modelling

UML Class Diagram & Entity Descriptions

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Prepared by: Project Team — End of Year Project

Status: Draft — Design Phase

1. Introduction

This document presents the Stage C Domain Modelling output for the CBC Digital Learning Platform. It builds directly on the functional requirements in the SRS (v1.0) and the use cases locked in Stage B to identify and formally describe the core domain entities of the system — the real-world concepts the platform must represent and manage.

Domain modelling at this stage captures what entities exist, what they know (attributes), and how they relate to each other (associations, compositions, and inheritance). This is distinct from database schema design (Stage F), which will derive from this model.

1.1 Modelling Decisions

Decision	Ruling	Rationale
User hierarchy	Learner, Teacher, MoESOfficial are subclasses of abstract User	Consistent with Stage B actor generalisation. Shared identity attributes (username, passwordHash, role) belong to User. Specialised attributes are pushed down to subclasses.
Content hierarchy	Lesson and TeacherMaterial are subclasses of abstract Content	Both share title, description, timestamps, and CBC tagging (Subject, ClassLevel, Competency). Lesson adds NCDC provenance; TeacherMaterial adds teacher ownership. Standard generalisation pattern.
Subject / ClassLevel / Competency	Three separate entities with associations	Supports independent querying required by FR-M-03 analytics. Reflects actual CBC structure where competencies are defined per subject per level.
AnalyticsDashboard	Removed — modelled as a derived view	Has no persistent state. Aggregates CompetencyProgress, AISession, and Post data at query time. Noted in the diagram.
Reaction vs Comment	Reaction holds type Enum {LIKE, SHARE}; Comment is a separate entity	Comment carries its own content attribute and authorship — structurally distinct from a boolean reaction. Reaction is a simple typed association between Learner and Post.

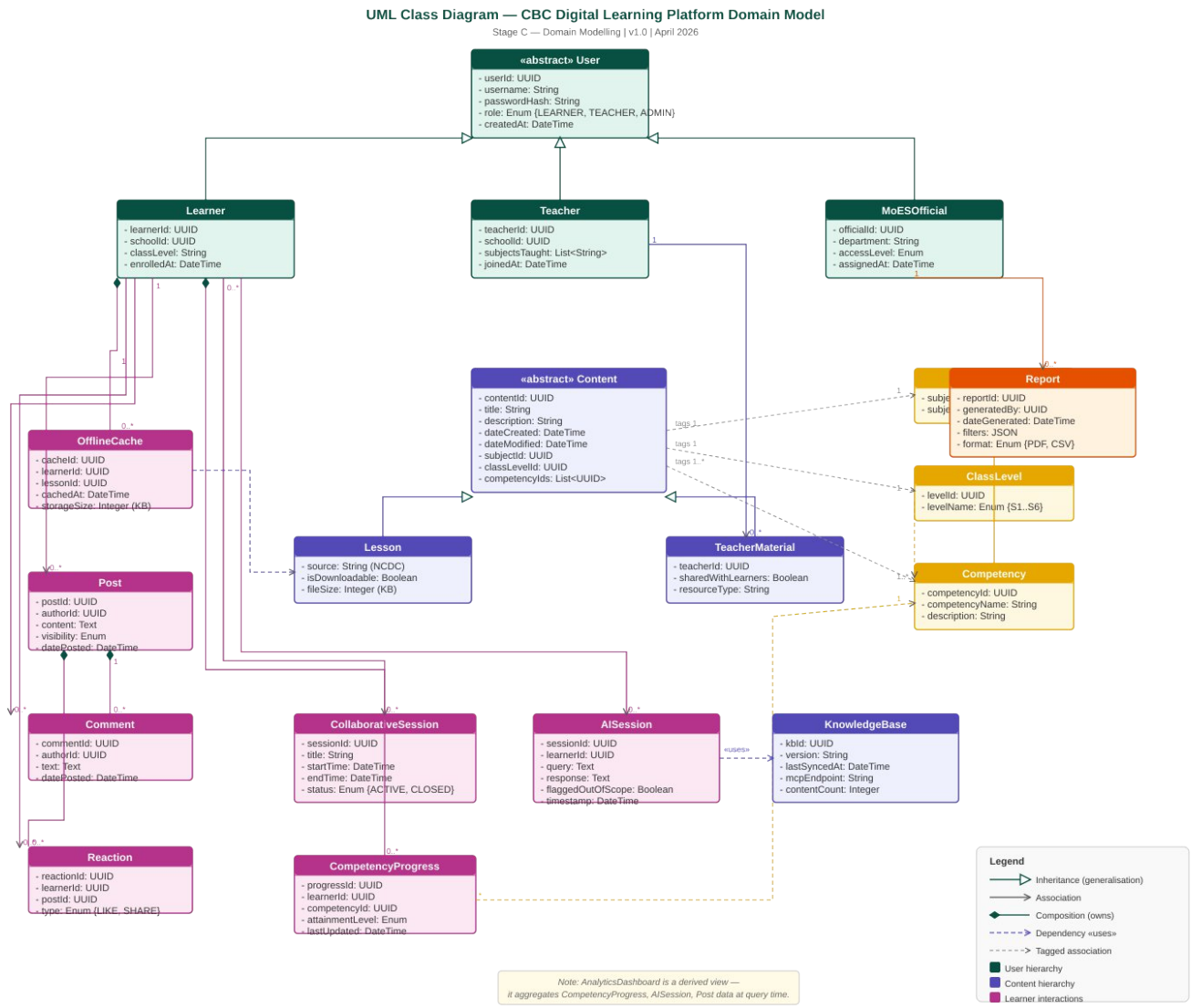
1.2 Entity Summary

#	Entity	Layer	Superclass	Derived From
1	User	User Hierarchy	—	FR-L-01, FR-T-01, FR-M-01
2	Learner	User Hierarchy	User	FR-L series
3	Teacher	User Hierarchy	User	FR-T series

#	Entity	Layer	Superclass	Derived From
4	MoESOfficial	User Hierarchy	User	FR-M series
5	Content	Content Hierarchy	—	FR-L-03, FR-T-02
6	Lesson	Content Hierarchy	Content	FR-L-03, FR-L-04
7	TeacherMaterial	Content Hierarchy	Content	FR-T-02, FR-T-03
8	Subject	CBC Structure	—	FR-L-03, FR-T-03
9	ClassLevel	CBC Structure	—	FR-L-03, FR-T-03
10	Competency	CBC Structure	—	FR-L-03, FR-L-09, FR-T-03
11	OfflineCache	Learner Interactions	—	FR-L-04, NFR-O-01
12	Post	Learner Interactions	—	FR-L-06, FR-L-07
13	Comment	Learner Interactions	—	FR-L-07 (UC-L-10)
14	Reaction	Learner Interactions	—	FR-L-07 (UC-L-10)
15	CollaborativeSession	Learner Interactions	—	FR-L-08
16	CompetencyProgress	Learner Interactions	—	FR-L-09
17	AISession	Learner Interactions	—	FR-L-05, FR-AI-01–05
18	KnowledgeBase	AI / MoES	—	FR-AI-03
19	Report	AI / MoES	—	FR-M-04

2. UML Class Diagram

The diagram below presents the complete domain model for the CBC Digital Learning Platform. Entities are organised in four layers: User hierarchy (top), Content hierarchy and CBC structure (middle), Learner interaction entities (lower), and MoES/AI entities (bottom right).



3. Entity Descriptions

Each entity below is described with its full attribute list, data types, and all relationships to other domain entities.

3.1 User Hierarchy

User «abstract»

Field	Detail
Entity Name	User
Stereotype	abstract
Superclass	—
Derived From	FR-L-01, FR-T-01, FR-M-01

Attributes:

Attribute	Type	Description
userId	UUID	Unique system-wide identifier for every user account
username	String	Unique login name chosen at registration
passwordHash	String	Bcrypt-hashed password — plain text never stored (NFR-SEC-03)
role	Enum {LEARNER, TEACHER, ADMIN}	Determines access permissions and dashboard routing
createdAt	DateTime	Timestamp of account creation

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Generalisation (subclass)	1 : 0..1	User is specialised as Learner
Teacher	Generalisation (subclass)	1 : 0..1	User is specialised as Teacher
MoESOfficial	Generalisation (subclass)	1 : 0..1	User is specialised as MoESOfficial

Learner

Field	Detail
Entity Name	Learner
Stereotype	—
Superclass	User
Derived From	FR-L series

Attributes:

Attribute	Type	Description
learnerId	UUID	Foreign key reference to User.userId
schoolId	UUID	Reference to the learner's registered school
classLevel	String	Current class level (S1–S6)
enrolledAt	DateTime	Date the learner was registered on the platform

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
User	Inheritance	1 : 1	Inherits all User attributes and authentication behaviour
OfflineCache	Composition	1 : 0..*	Learner owns their offline cache records
Post	Association	1 : 0..*	Learner authors zero or many Posts
Comment	Association	1 : 0..*	Learner authors zero or many Comments
Reaction	Association	1 : 0..*	Learner makes zero or many Reactions
CollaborativeSession	Association (M:M)	0..* : 0..*	Learner joins zero or many Sessions
CompetencyProgress	Composition	1 : 0..*	Learner owns their progress records
AISession	Association	1 : 0..*	Learner initiates zero or many AI Sessions

Teacher

Field	Detail
Entity Name	Teacher
Stereotype	—
Superclass	User
Derived From	FR-T series

Attributes:

Attribute	Type	Description
teacherId	UUID	Foreign key reference to User.userId
schoolId	UUID	Reference to the teacher's registered school
subjectsTaught	List<String>	List of CBC subject names the teacher is responsible for
joinedAt	DateTime	Date the teacher was registered on the platform

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
User	Inheritance	1 : 1	Inherits all User attributes and authentication behaviour
TeacherMaterial	Association	1 : 0..*	Teacher authors zero or many TeacherMaterials

MoESOfficial

Field	Detail
Entity Name	MoESOfficial
Stereotype	—
Superclass	User
Derived From	FR-M series

Attributes:

Attribute	Type	Description
officialId	UUID	Foreign key reference to User.userId
department	String	Ministry department or NCDC division

Attribute	Type	Description
accessLevel	Enum	Governance permission tier controlling dashboard and content access
assignedAt	DateTime	Date the official was provisioned on the platform

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
User	Inheritance	1 : 1	Inherits all User attributes and authentication behaviour
Report	Association	1 : 0..*	Official generates zero or many Reports

3.2 Content Hierarchy

Content «abstract»

Field	Detail
Entity Name	Content
Stereotype	abstract
Superclass	—
Derived From	FR-L-03, FR-T-02, FR-T-03

Attributes:

Attribute	Type	Description
contentId	UUID	Unique identifier for all content items
title	String	Display title of the content
description	String	Brief summary of the content's learning objective
dateCreated	DateTime	Timestamp of initial creation
dateModified	DateTime	Timestamp of last modification
subjectId	UUID	Reference to the associated Subject
classLevelId	UUID	Reference to the associated ClassLevel
competencyIds	List<UUID>	References to one or more associated Competencies

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Lesson	Generalisation (subclass)	1 : 0..*	Content is specialised as NCDC-authorised Lesson
TeacherMaterial	Generalisation (subclass)	1 : 0..*	Content is specialised as teacher-contributed material
Subject	Association	1 : 1	Content is tagged to exactly one Subject
ClassLevel	Association	1 : 1	Content is tagged to exactly one ClassLevel
Competency	Association	1 : 1..*	Content is tagged to one or more Competencies

Lesson

Field	Detail
Entity Name	Lesson
Stereotype	—
Superclass	Content
Derived From	FR-L-03, FR-L-04

Attributes:

Attribute	Type	Description
source	String	Content authority — always 'NCDC' for curriculum lessons
isDownloadable	Boolean	Flag indicating whether the lesson can be cached for offline access
fileSize	Integer (KB)	Compressed size used for storage management in OfflineCache

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Content	Inheritance	1 : 1	Inherits all Content attributes and CBC tagging
OfflineCache	Association	1 : 0..*	Lesson may exist in zero or many learner offline caches

TeacherMaterial

Field	Detail
Entity Name	TeacherMaterial
Stereotype	—
Superclass	Content
Derived From	FR-T-02, FR-T-03, FR-T-04

Attributes:

Attribute	Type	Description
teacherId	UUID	Reference to the authoring Teacher
sharedWithLearners	Boolean	Flag controlling whether the material is visible

Attribute	Type	Description
		to learners
resourceType	String	Descriptor of material type (e.g., study guide, past paper, notes)

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Content	Inheritance	1 : 1	Inherits all Content attributes and CBC tagging
Teacher	Association	0..* : 1	Many TeacherMaterials belong to one Teacher author

3.3 CBC Structure Entities

Subject

Field	Detail
Entity Name	Subject
Stereotype	—
Superclass	—
Derived From	FR-L-03, FR-T-03

Attributes:

Attribute	Type	Description
subjectId	UUID	Unique identifier for the subject
subjectName	String	Official CBC subject name (e.g., Mathematics, Biology)

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Competency	Composition	1 : 1..*	Subject owns one or many Competencies — competency cannot exist without a subject
Content	Association	1 : 0..*	Subject is referenced by zero or many content items

ClassLevel

Field	Detail
Entity Name	ClassLevel
Stereotype	—
Superclass	—
Derived From	FR-L-03, FR-T-03

Attributes:

Attribute	Type	Description
levelId	UUID	Unique identifier for the class level

Attribute	Type	Description
levelName	Enum {S1, S2, S3, S4, S5, S6}	Uganda secondary school year group

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Competency	Association	1 : 0..*	ClassLevel is associated with zero or many Competencies
Content	Association	1 : 0..*	ClassLevel is referenced by zero or many content items

Competency

Field	Detail
Entity Name	Competency
Stereotype	—
Superclass	—
Derived From	FR-L-03, FR-L-09, FR-T-03

Attributes:

Attribute	Type	Description
competencyId	UUID	Unique identifier for the competency
competencyName	String	Official CBC competency name
description	String	Detailed description of the expected learning outcome

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Subject	Composition (child)	0..* : 1	Competency belongs to exactly one Subject
ClassLevel	Association	0..* : 1	Competency is targeted at exactly one ClassLevel
Content	Association	0..* : 1..*	Competency is referenced by one or many content items
CompetencyProgress	Association	1 : 0..*	Competency is tracked by zero or many progress records

3.4 Learner Interaction Entities

OfflineCache

Field	Detail
Entity Name	OfflineCache
Stereotype	—
Superclass	—
Derived From	FR-L-04, NFR-O-01

Attributes:

Attribute	Type	Description
cacheId	UUID	Unique identifier for the cache record
learnerId	UUID	Reference to the owning Learner
lessonId	UUID	Reference to the cached Lesson
cachedAt	DateTime	Timestamp when the lesson was downloaded
storageSize	Integer (KB)	Size of the cached lesson on the device

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Composition (child)	0..* : 1	Cache record belongs to exactly one Learner
Lesson	Association	0..* : 1	Cache record references exactly one Lesson

Post

Field	Detail
Entity Name	Post
Stereotype	—
Superclass	—
Derived From	FR-L-06, FR-L-07, NFR-SEC-05

Attributes:

Attribute	Type	Description
postId	UUID	Unique identifier for the post
authorId	UUID	Reference to the authoring Learner
content	Text	The body text of the learner's insight or research contribution
visibility	Enum {PRIVATE, PEERS, TEACHERS, PUBLIC}	Access control level for the post (private by default)
datePosted	DateTime	Timestamp of post creation

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Association	0..* : 1	Post is authored by exactly one Learner
Comment	Composition	1 : 0..*	Post owns zero or many Comments
Reaction	Composition	1 : 0..*	Post owns zero or many Reactions

Comment

Field	Detail
Entity Name	Comment
Stereotype	—
Superclass	—
Derived From	FR-L-07 (UC-L-10)

Attributes:

Attribute	Type	Description
commentId	UUID	Unique identifier for the comment
authorId	UUID	Reference to the commenting Learner
text	Text	The body text of the comment
datePosted	DateTime	Timestamp of comment creation

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Post	Composition (child)	0..* : 1	Comment belongs to exactly one Post
Learner	Association	0..* : 1	Comment is authored by exactly one Learner

Reaction

Field	Detail
Entity Name	Reaction
Stereotype	—
Superclass	—
Derived From	FR-L-07 (UC-L-10)

Attributes:

Attribute	Type	Description
reactionId	UUID	Unique identifier for the reaction
learnerId	UUID	Reference to the reacting Learner
postId	UUID	Reference to the Post being reacted to
type	Enum {LIKE, SHARE}	The type of reaction made by the learner
dateReacted	DateTime	Timestamp of the reaction

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Post	Composition (child)	0..* : 1	Reaction belongs to exactly one Post
Learner	Association	0..* : 1	Reaction is made by exactly one Learner

CollaborativeSession

Field	Detail
Entity Name	CollaborativeSession
Stereotype	—

Field	Detail
Superclass	—
Derived From	FR-L-08

Attributes:

Attribute	Type	Description
sessionId	UUID	Unique identifier for the session
title	String	Display name of the collaborative session
startTime	DateTime	Scheduled or actual start time
endTime	DateTime	Actual end time (null while session is active)
status	Enum {ACTIVE, CLOSED}	Current state of the session

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Association (M:M)	0..* : 0..*	Many learners can join many sessions

CompetencyProgress

Field	Detail
Entity Name	CompetencyProgress
Stereotype	—
Superclass	—
Derived From	FR-L-09

Attributes:

Attribute	Type	Description
progressId	UUID	Unique identifier for the progress record
learnerId	UUID	Reference to the owning Learner
competencyId	UUID	Reference to the tracked Competency
attainmentLevel	Enum {NOT_STARTED, IN_PROGRESS, ATTAINED}	Learner's current attainment status for this competency

Attribute	Type	Description
lastUpdated	DateTime	Timestamp of last attainment update

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Composition (child)	0..* : 1	Progress record belongs to exactly one Learner
Competency	Association	0..* : 1	Progress record tracks exactly one Competency

AISession

Field	Detail
Entity Name	AISession
Stereotype	—
Superclass	—
Derived From	FR-L-05, FR-AI-01 to FR-AI-05

Attributes:

Attribute	Type	Description
sessionId	UUID	Unique identifier for the AI interaction record
learnerId	UUID	Reference to the initiating Learner
query	Text	The academic question submitted by the learner
response	Text	The AI Tutor's contextualised response
flaggedOutOfScope	Boolean	True if the query fell outside the Knowledge Base
timestamp	DateTime	Timestamp of the interaction

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
Learner	Association	0..* : 1	AI Session is initiated by exactly one Learner
KnowledgeBase	Dependency «uses»	0..* : 1	AI Session queries the KnowledgeBase via MCP to generate responses

3.5 MoES / AI Entities

KnowledgeBase

Field	Detail
Entity Name	KnowledgeBase
Stereotype	—
Superclass	—
Derived From	FR-AI-03, FR-AI-02

Attributes:

Attribute	Type	Description
kbId	UUID	Unique identifier for the knowledge base instance
version	String	Current version of the knowledge base content
lastSyncedAt	DateTime	Timestamp of the most recent MCP content sync
mcpEndpoint	String	URL of the MCP server endpoint used by the AI Tutor
contentCount	Integer	Total number of content records indexed in the knowledge base

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
AISession	Dependency «uses»	1 : 0..*	KnowledgeBase is queried by zero or many AISession interactions

Report

Field	Detail
Entity Name	Report
Stereotype	—
Superclass	—
Derived From	FR-M-04

Attributes:

Attribute	Type	Description
reportId	UUID	Unique identifier for the generated report
generatedBy	UUID	Reference to the MoESOfficial who generated the report
dateGenerated	DateTime	Timestamp of report generation
filters	JSON	Serialised filter parameters (date range, region, subject, school)
format	Enum {PDF, CSV}	Export format selected at generation time

Relationships:

Related Entity	Relationship Type	Multiplicity	Description
MoESOfficial	Association	0..* : 1	Report is generated by exactly one MoESOfficial

4. Requirements Traceability — Stage C

The table below confirms that every domain entity traces back to at least one functional or non-functional requirement in the SRS.

Entity	SRS Requirement(s)	Stage B Use Case(s)
User	FR-L-01, FR-T-01, FR-M-01, NFR-SEC-03	UC-U-01, UC-U-02, UC-U-03
Learner	FR-L-01 to FR-L-11	UC-L-04 to UC-L-13
Teacher	FR-T-01 to FR-T-04	UC-T-04 to UC-T-09
MoESOfficial	FR-M-01 to FR-M-04	UC-M-03 to UC-M-08
Content	FR-L-03, FR-T-02, FR-T-03	UC-L-04, UC-T-04, UC-T-05
Lesson	FR-L-03, FR-L-04	UC-L-04, UC-L-05, UC-L-06
TeacherMaterial	FR-T-02, FR-T-03, FR-T-04	UC-T-04, UC-T-05, UC-T-06
Subject	FR-L-03, FR-T-03	UC-L-04, UC-T-05
ClassLevel	FR-L-03, FR-T-03	UC-L-04, UC-T-05
Competency	FR-L-03, FR-L-09, FR-T-03	UC-L-04, UC-L-12, UC-T-05
OfflineCache	FR-L-04, NFR-O-01	UC-L-05, UC-L-06
Post	FR-L-06, FR-L-07, NFR-SEC-05	UC-L-08, UC-L-09, UC-L-10
Comment	FR-L-07	UC-L-10
Reaction	FR-L-07	UC-L-10
CollaborativeSession	FR-L-08	UC-L-11
CompetencyProgress	FR-L-09	UC-L-12
AISession	FR-L-05, FR-AI-01 to FR-AI-05	UC-L-07, UC-AI-01 to UC-AI-05
KnowledgeBase	FR-AI-03	UC-AI-02
Report	FR-M-04	UC-M-07, UC-M-08